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Supplemental information

Circulating T cell profiles associate with enterotype signatures underlying hematological malignancy relapses

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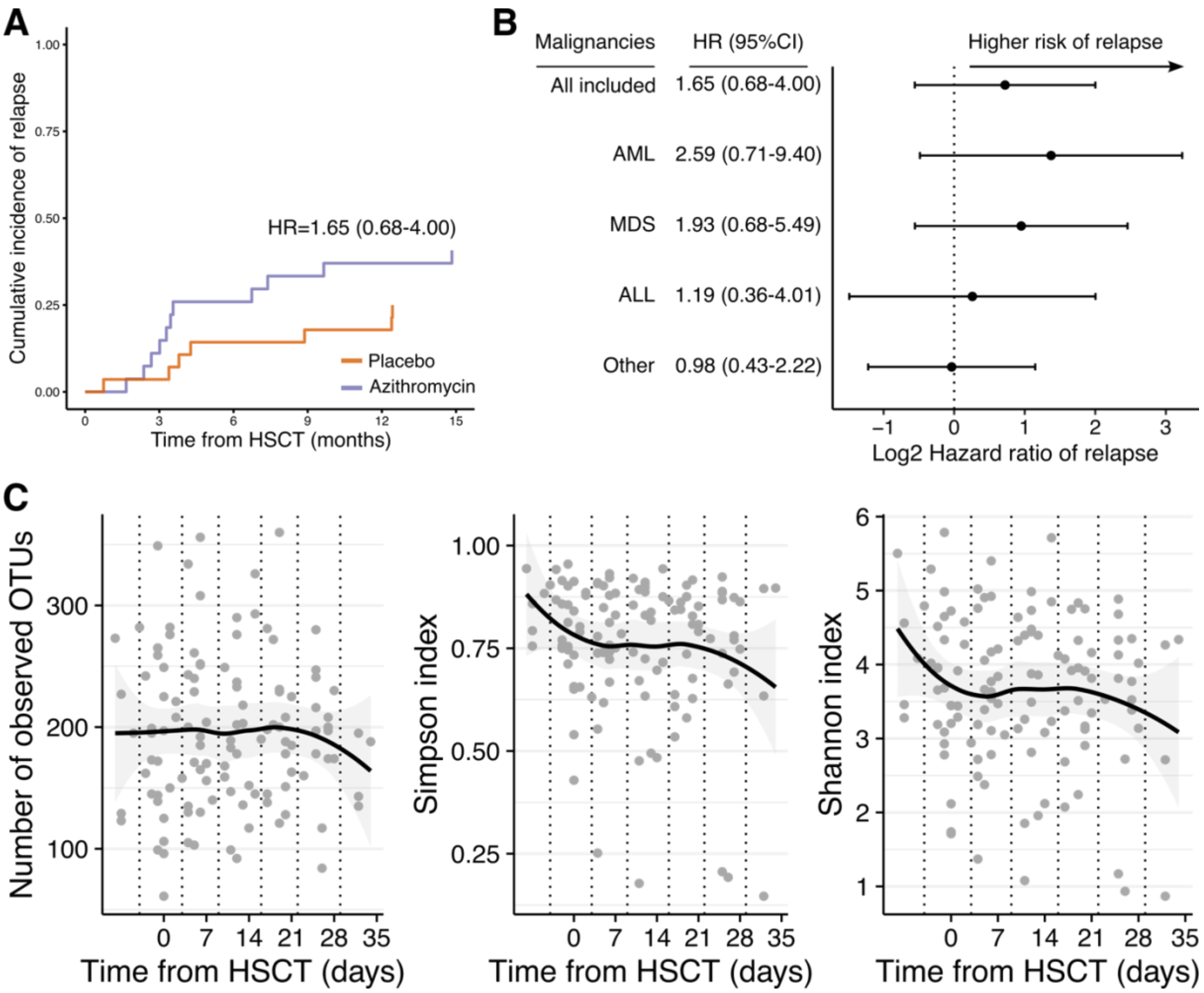
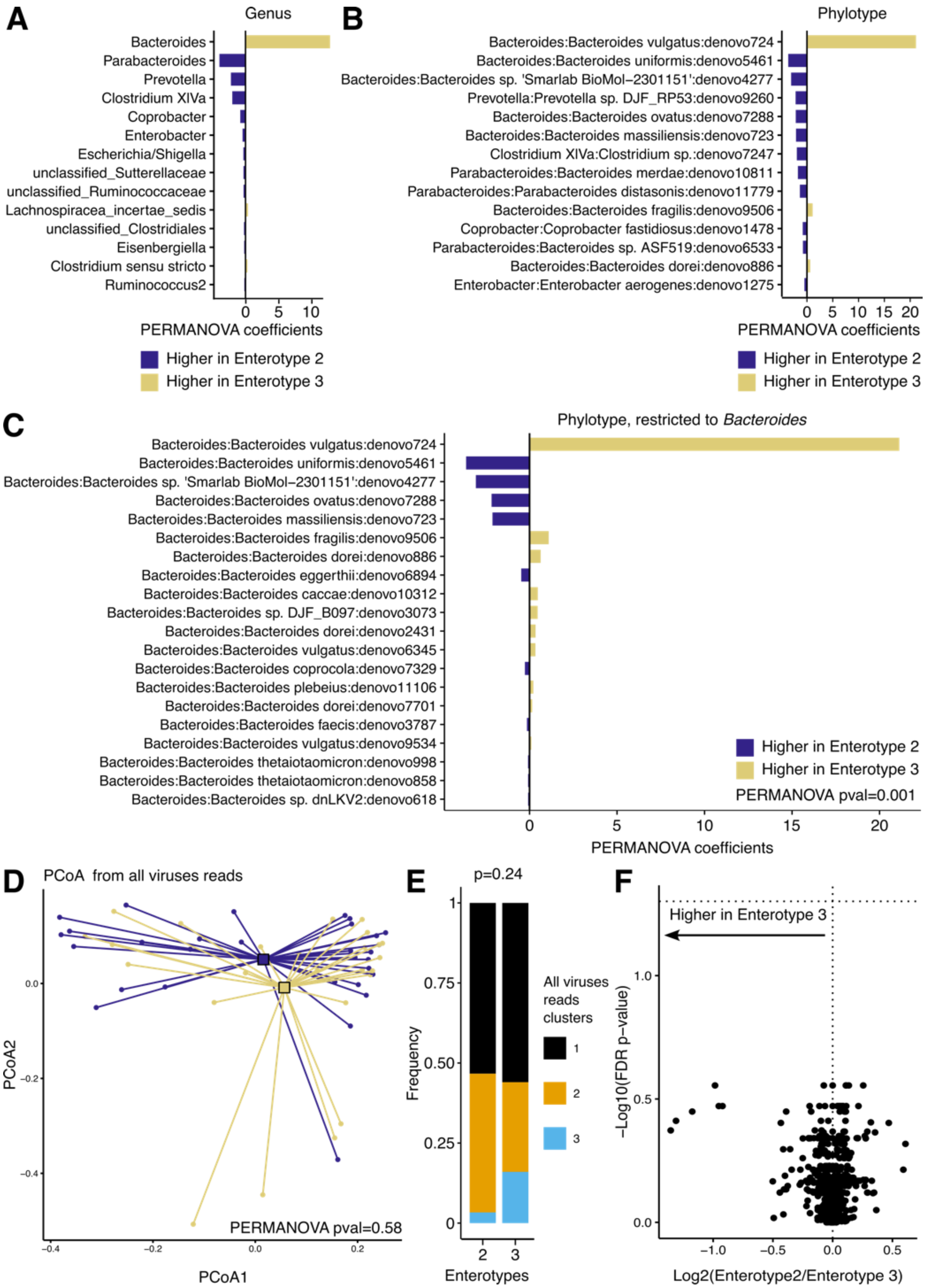


Figure S1. Relapse incidence and alpha-diversity indexes, related to Figure 1 and Figure 2. A.) Cumulative incidence of relapse with death defined as a competing risk computed with all included patients. B.) Subgroup analyses of relapse according to hematological malignancy. Using Fine and Gray model, death was defined as a competing risk. C.) Alpha-diversity indexes according to time from allogeneic hematopoietic stem cell transplantation (HSCT). Dot plots and regression line with 95% confidence interval are shown. OTU: operational taxonomic unit.



A
Genus
Bacteroides
Parabacteroides
Prevotella
Clostridium XlVa
Coprobacter
Enterobacter
Escherichia/Shigella
unclassified_Sutterellaceae
unclassified_Ruminococcaceae
Lachnospiracea_incertae_sedis
unclassified_Clostridiales
Eisenbergiella
Clostridium sensu stricto
Ruminococcus2
PERMANOVA coefficients
Higher in Enterotype 2
Higher in Enterotype 3
B
Phylotype
Bacteroides:Bacteroides vulgatus:denovo724
Bacteroides:Bacteroides uniformis:denovo5461
Bacteroides:Bacteroides sp. 'Smarlab BioMol−2301151':denovo4277
Prevotella:Prevotella sp. DJF_RP53:denovo9260
Bacteroides:Bacteroides ovatus:denovo7288
Bacteroides:Bacteroides massiliensis:denovo723
Clostridium XlVa:Clostridium sp.:denovo7247
Parabacteroides:Bacteroides merdae:denovo10811
Parabacteroides:Parabacteroides distasonis:denovo11779
Bacteroides:Bacteroides fragilis:denovo9506
Coprobacter:Coprobacter fastidiosus:denovo1478
Parabacteroides:Bacteroides sp. ASF519:denovo6533
Bacteroides:Bacteroides dorei:denovo886
Enterobacter:Enterobacter aerogenes:denovo1275
PERMANOVA coefficients
Higher in Enterotype 2
Higher in Enterotype 3
C
Phylotype, restricted to *Bacteroides*
Bacteroides:Bacteroides vulgatus:denovo724
Bacteroides:Bacteroides uniformis:denovo5461
Bacteroides:Bacteroides sp. 'Smarlab BioMol−2301151':denovo4277
Bacteroides:Bacteroides ovatus:denovo7288
Bacteroides:Bacteroides massiliensis:denovo723
Bacteroides:Bacteroides fragilis:denovo9506
Bacteroides:Bacteroides dorei:denovo886
Bacteroides:Bacteroides eggerthii:denovo6894
Bacteroides:Bacteroides caccae:denovo10312
Bacteroides:Bacteroides sp. DJF_B097:denovo3073
Bacteroides:Bacteroides dorei:denovo2431
Bacteroides:Bacteroides vulgatus:denovo6345
Bacteroides:Bacteroides coprocola:denovo7329
Bacteroides:Bacteroides plebeius:denovo11106
Bacteroides:Bacteroides dorei:denovo7701
Bacteroides:Bacteroides faecis:denovo3787
Bacteroides:Bacteroides vulgatus:denovo9534
Bacteroides:Bacteroides thetaiotaomicron:denovo998
Bacteroides:Bacteroides thetaiotaomicron:denovo858
Bacteroides:Bacteroides sp. dnLKV2:denovo618
Higher in Enterotype 2
Higher in Enterotype 3
PERMANOVA pval=0.001
PERMANOVA coefficients
D
PCoA from all viruses reads
PCoA2
PCoA1
PERMANOVA pval=0.58
E
p=0.24
Frequency
All viruses reads clusters
1
2
3
Enterotypes
F
Higher in Enterotype 3
−Log10(FDR p−value)
Log2(Enterotype2/Enterotype 3)

Figure S2. Comparison of the two Bacteroides enriched enterotypes (2 and 3) gut viruses and metabolites profiles, related to Figure 2. A.) Permutational multivariate analysis of variance (PERMANOVA) coefficients when comparing genus abundances from enterotypes 2 and 3. B.) PERMANOVA coefficients when comparing phylotypes abundances from enterotypes 2 and 3. C.) PERMANOVA coefficients when comparing Bacteroides phylotypes from enterotypes 2 and 3. D.) Principal co-ordinate analysis (PCoA) with UniFrac distance matrix from all viruses. Samples are colored according to their respective bacterial enterotypes 2 or enterotype 3. PERMANOVA analysis did not revealed significant differences. E.) Bar plot showing frequencies of viral enterotypes according to bacterial enterotypes 2 and enterotype 3. Fisher exact test was performed to compute P-values. F.) Volcano plot showing differences in metabolites levels between enterotype 2 and enterotype 3. Non-parametric Wilcoxon test was performed and adjusted for false discovery rate (Benjamini Hochberg).

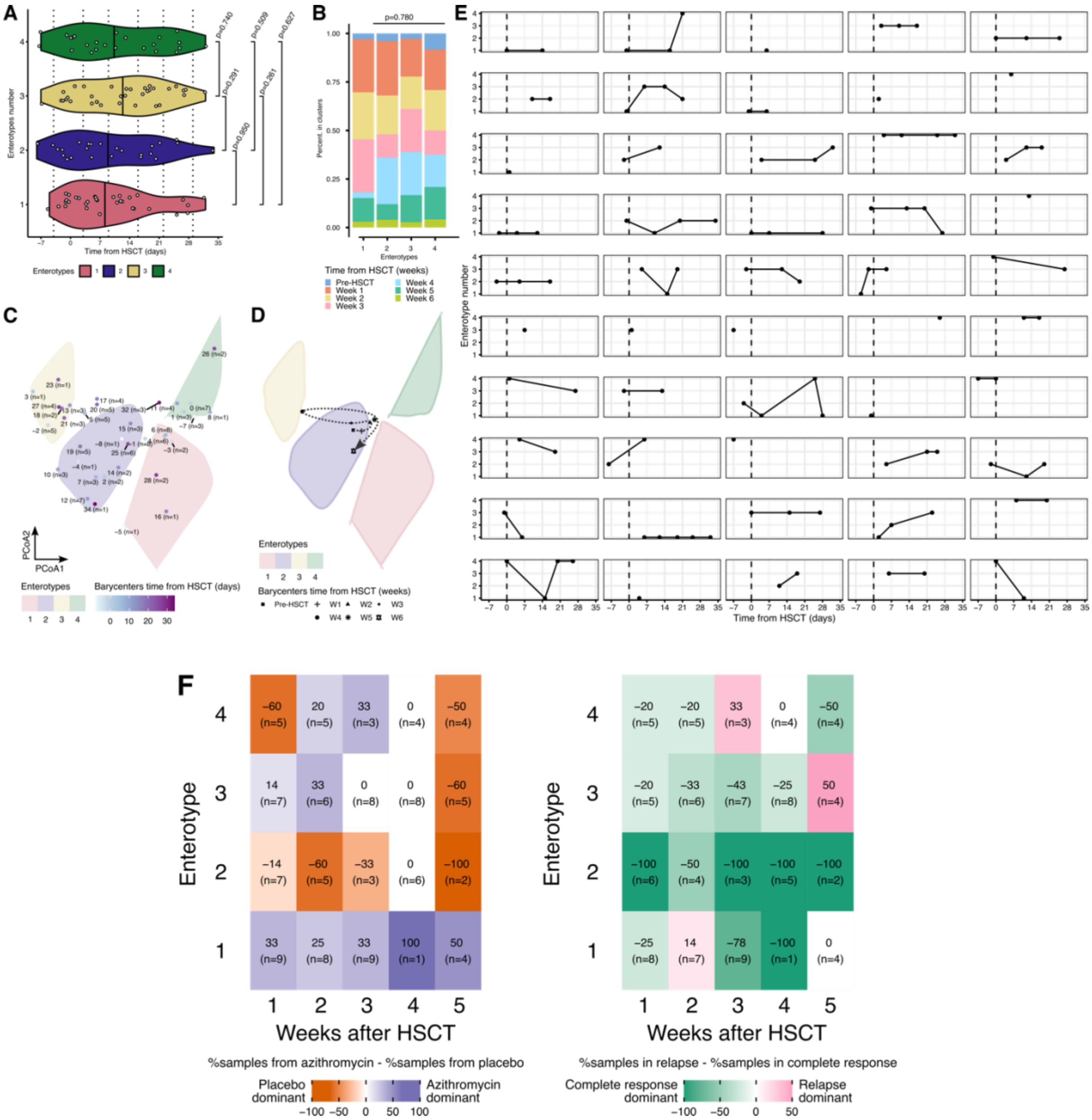
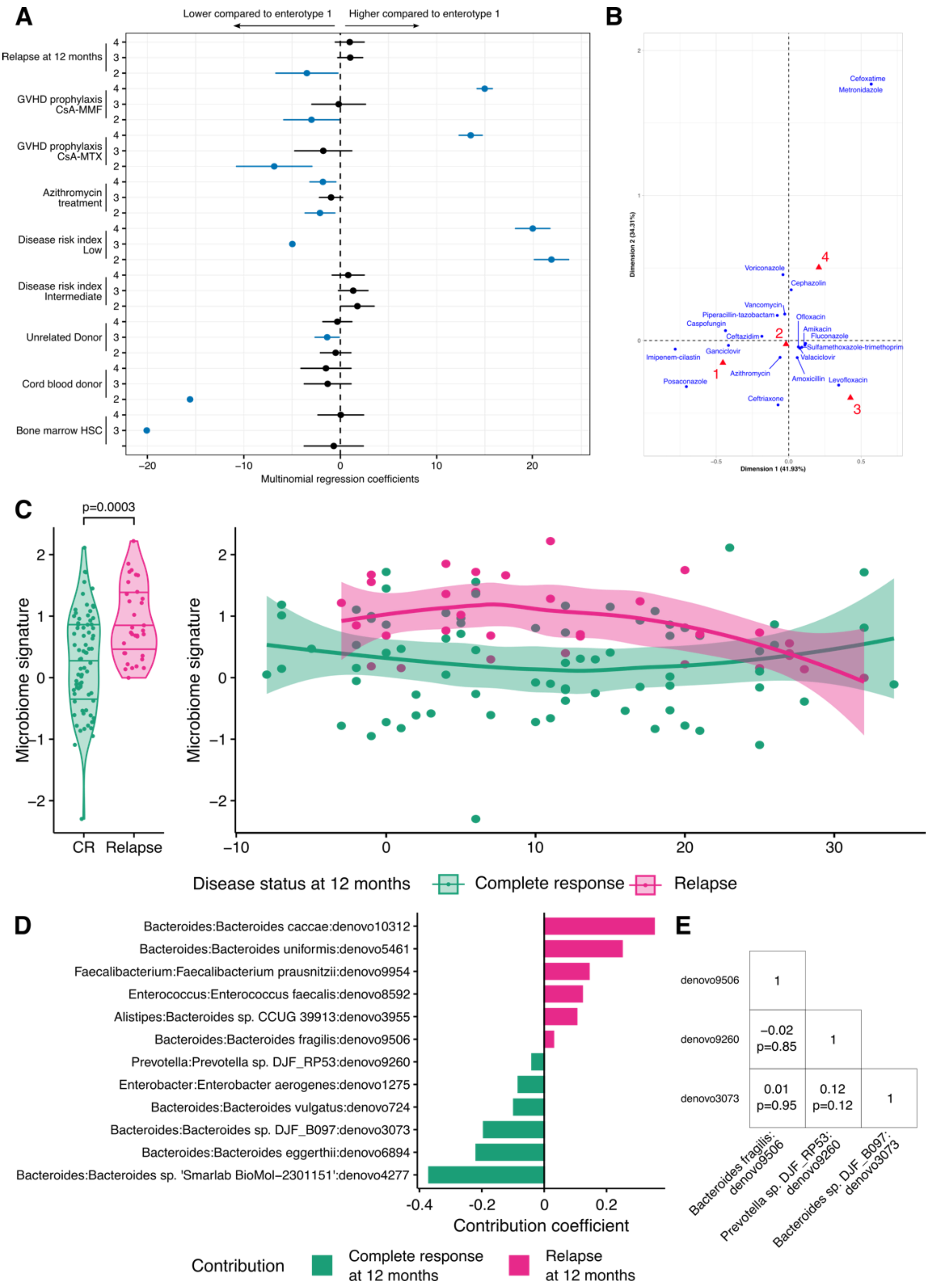


Figure S3. Enterotype according to time from allogeneic hematopoietic stem cell transplantation, related to Figure 3. A.) Comparison of time from allogeneic hematopoietic stem cell transplantation (allo-HSCT) between enterotypes, with time as a continuous variable. Dotted lines show weeks defined in Figure 1. P-value was computed with bilateral Wilcoxon test. B.) Comparison of time from allo-HSCT between enterotypes with time as a categorical variable. Weeks are defined according to sample distribution shown in Figure 1. P-value was computed with Fisher's exact test. C.) Barycenters of time from allo-HSCT and enterotypes clusters areas. Dots represents the barycenter of a day from allo-HSCT. Numbers describe the days from HSCT with the corresponding number of samples. D.) Barycenter and trajectory of samples from weeks after a HSCT as defined in Figure 1. E.) Patients individual enterotype trajectories according to time from transplantation. Each plot shows one patient sample according to time from allogeneic hematopoietic stem cell transplantation (HSCT). Vertical dashed line represents day of graft infusion. F.) Heatmaps depicting frequencies of samples in group of randomizations / complete response or relapse at 12 months according to enterotypes across time. Weeks are defined according to sample distribution shown in Figure 1.



A
Lower compared to enterotype 1
Higher compared to enterotype 1
Relapse at 12 months
GVHD prophylaxis CsA-MMF
GVHD prophylaxis CsA-MTX
Azithromycin treatment
Disease risk index Low
Disease risk index Intermediate
Unrelated Donor
Cord blood donor
Bone marrow HSC
Multinomial regression coefficients
B
Dimension 1 (41.93%)
Dimension 2 (34.31%)
Cefoxatime
Metronidazole
Voriconazole
Cephazolin
Vancomycin
Piperacillin-tazobactam
Caspofungin
Ofloxacin
Amikacin
Fluconazole
Ceftazidim
Sulfamethoxazole-trimethoprim
Ganciclovir
Imipenem-cilastin
Valaciclovir
Azithromycin
Amoxicillin
Levofloxacin
Posaconazole
Ceftriaxone
C
p=0.0003
Microbiome signature
CR
Relapse
Disease status at 12 months
Complete response
Relapse
D
Bacteroides:Bacteroides caccae:denovo10312
Bacteroides:Bacteroides uniformis:denovo5461
Faecalibacterium:Faecalibacterium prausnitzii:denovo9954
Enterococcus:Enterococcus faecalis:denovo8592
Alistipes:Bacteroides sp. CCUG 39913:denovo3955
Bacteroides:Bacteroides fragilis:denovo9506
Prevotella:Prevotella sp. DJF_RP53:denovo9260
Enterobacter:Enterobacter aerogenes:denovo1275
Bacteroides:Bacteroides vulgatus:denovo724
Bacteroides:Bacteroides sp. DJF_B097:denovo3073
Bacteroides:Bacteroides eggerthii:denovo6894
Bacteroides:Bacteroides sp. 'Smarlab BioMol−2301151':denovo4277
Contribution coefficient
Contribution
Complete response at 12 months
Relapse at 12 months
E
denovo9506
denovo9260
denovo3073
1
−0.02 p=0.85
1
0.01 p=0.95
0.12 p=0.12
1
Bacteroides fragilis: denovo9506
Prevotella sp. DJF_RP53: denovo9260
Bacteroides sp. DJF_B097: denovo3073

Figure S4. Characteristics of gut microbiota associated with relapse, related to Figure 3. A.) Multivariate multinomial regression models evaluating enterotypes and clinical variable association. Multivariate model includes variables associated with enterotypes (Table 1). Blue dots and line show statistically significant co-efficients. Coefficients are relative to enterotype 1 and unmentioned group in each variable. GVHD: graft-versus-host disease, CsA: cyclosporine A, MMF: mycophenolate mofetil, MTX: methotrexate, HSC: hematopoietic stem cell. B.) Correspondence analysis map of antibiotics used in the four enterotypes. Blue dots show barycenter of antibiotics used and red triangles the barycenter of enterotypes. Each barycenter represents the co-ordinates of mean distribution of the used antibiotic among the enterotypes. C.) Dynamic microbial signature associated with relapse at 12 months. Violin plot and regression curve with 95%CI showing microbial signature in samples associated with relapse and complete responses. D.) Bar plot describing taxa contribution to microbial signature. E.) Correlation analysis between phylotypes associated with relapse. Spearman co-efficients and p-value adjusted with Benjamini-Hochberg method are shown.

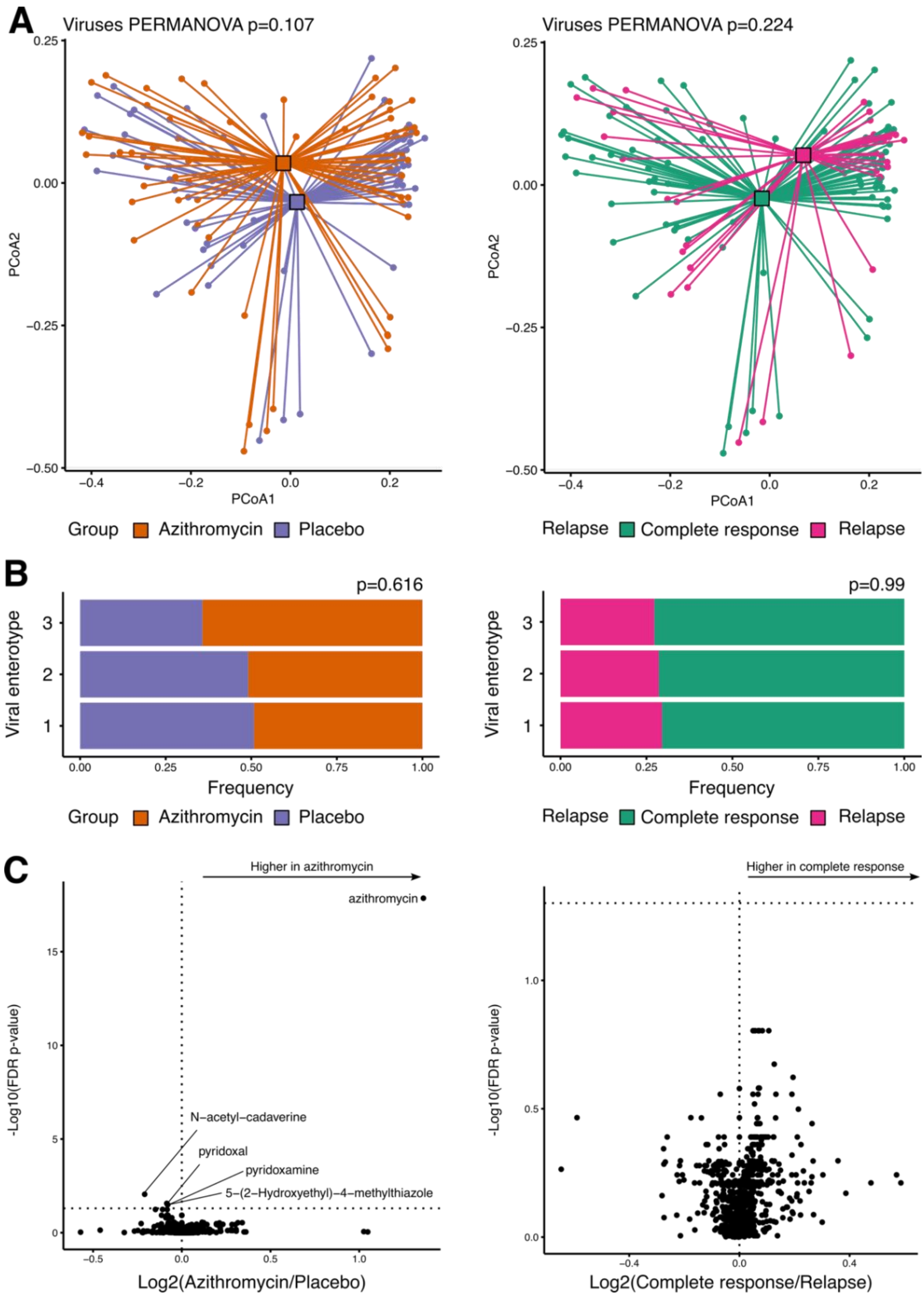

A
Viruses PERMANOVA p=0.107
PCoA2
PCoA1
Group Azithromycin Placebo
Viruses PERMANOVA p=0.224
PCoA2
PCoA1
Relapse Complete response Relapse
B
p=0.616
Viral enterotype
Frequency
Group Azithromycin Placebo
p=0.99
Viral enterotype
Frequency
Relapse Complete response Relapse
C
Higher in azithromycin
azithromycin
N−acetyl−cadaverine
pyridoxal
pyridoxamine
5−(2−Hydroxyethyl)−4−methylthiazole
-Log10(FDR p-value)
Log2(Azithromycin/Placebo)
Higher in complete response
-Log10(FDR p-value)
Log2(Complete response/Relapse)

Figure S5. Comparison of gut viruses' composition and gut metabolites levels between patients in azithromycin and placebo groups and between patients in relapse or complete responses, related to Figure 3. A.) Principal co-ordinate analysis (PCoA) with UniFrac distance matrix from all viruses. Samples are colored according to azithromycin or placebo groups or according to relapse or complete responses statuses. Permutational multivariate analysis of variance (PERMANOVA) analyses did not revealed significant differences. B.) Bar plots describing frequencies of viral enterotypes according to azithromycin or placebo groups or according to relapse or complete responses statuses. Fisher exact test was performed to compute P-values. C.) Volcano plot showing differences in metabolites levels between azithromycin and placebo groups and between relapse and complete responses statuses. Non-parametric Wilcoxon test was performed and adjusted for false discovery rate (Benjamini Hochberg).

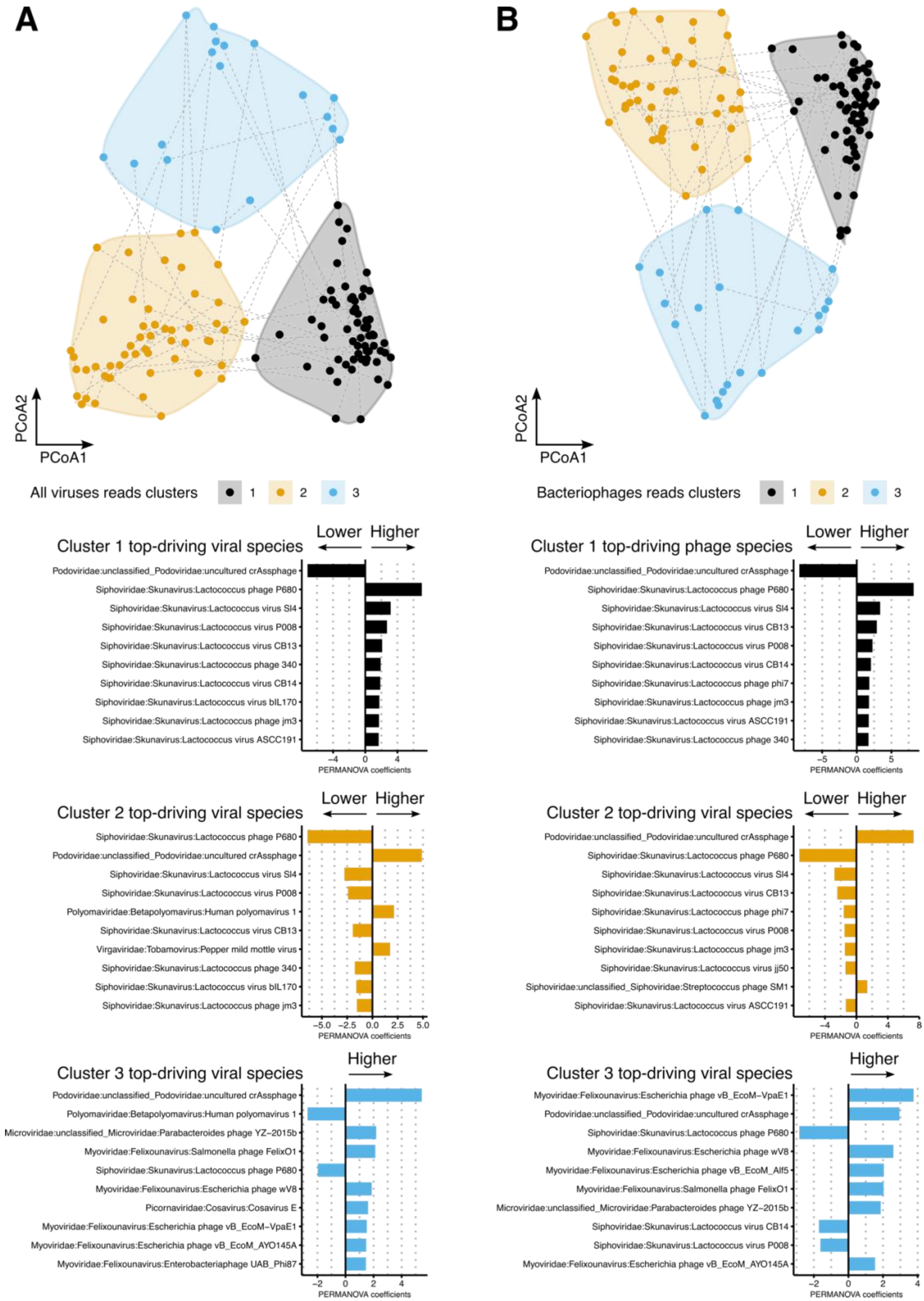

A
B
PCoA2
PCoA1
All viruses reads clusters 1 2 3
Bacteriophages reads clusters 1 2 3
Cluster 1 top-driving viral species
Cluster 1 top-driving phage species
Cluster 2 top-driving viral species
Cluster 3 top-driving viral species
Lower
Higher
PERMANOVA coefficients
Podoviridae:unclassified_Podoviridae:uncultured crAssphage
Siphoviridae:Skunavirus:Lactococcus phage P680
Siphoviridae:Skunavirus:Lactococcus virus Sl4
Siphoviridae:Skunavirus:Lactococcus virus P008
Siphoviridae:Skunavirus:Lactococcus virus CB13
Siphoviridae:Skunavirus:Lactococcus phage 340
Siphoviridae:Skunavirus:Lactococcus virus CB14
Siphoviridae:Skunavirus:Lactococcus virus bIL170
Siphoviridae:Skunavirus:Lactococcus phage jm3
Siphoviridae:Skunavirus:Lactococcus virus ASCC191
Siphoviridae:Skunavirus:Lactococcus phage phi7
Polyomaviridae:Betapolyomavirus:Human polyomavirus 1
Virgaviridae:Tobamovirus:Pepper mild mottle virus
Siphoviridae:Skunavirus:Lactococcus virus jj50
Siphoviridae:unclassified_Siphoviridae:Streptococcus phage SM1
Microviridae:unclassified_Microviridae:Parabacteroides phage YZ−2015b
Myoviridae:Felixounavirus:Salmonella phage FelixO1
Myoviridae:Felixounavirus:Escherichia phage wV8
Picornaviridae:Cosavirus:Cosavirus E
Myoviridae:Felixounavirus:Escherichia phage vB_EcoM−VpaE1
Myoviridae:Felixounavirus:Escherichia phage vB_EcoM_AYO145A
Myoviridae:Felixounavirus:Enterobacteriaphage UAB_Phi87
Myoviridae:Felixounavirus:Escherichia phage vB_EcoM_Alf5

Figure S6. Principal co-ordinate analysis of viral reads and viral clusters composition, related to Figure 4. A.) Principal co-ordinate analysis (PCoA) with UniFrac distance matrix from all viruses reads. B.) Principal co-ordinate analysis with UniFrac distance matrix from bacteriophages reads only. Samples were clustered with hierarchical K-means. Driving species were computed with permutational multivariate analysis of variance (PERMANOVA).

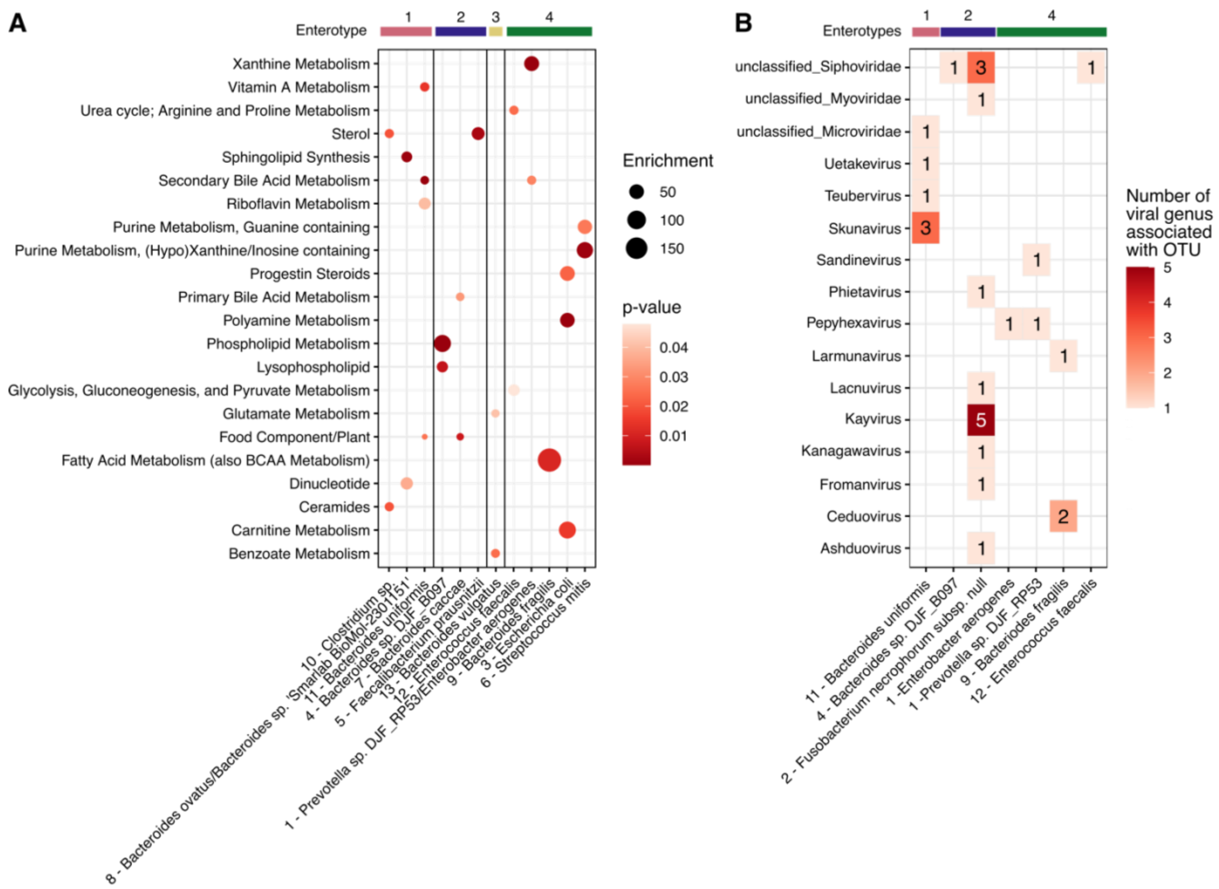


Figure S7. Metabolic pathways and viral genera are specifically associated with bacteria taxa, related to Figure 5. A.) Enriched pathways of metabolites correlated with network modules. Dot plot depicting statistically significant enriched pathways in network modules. Module numbers are associated with phylotypes species name. Phylotypes are grouped according to the enterotype they are driving (Figure S8) Enrichment was computed with over-representation method and P-value with hypergeometric test. B.) Viruses' genus associated with enterotypes top driving phylotypes. Tile plot showing number of species associated with phylotypes (OTUs) and which belongs to the same genus. Modules number are associated with phylotypes species name. Phylotypes are grouped according to the enterotype they are driving (Figure S8).

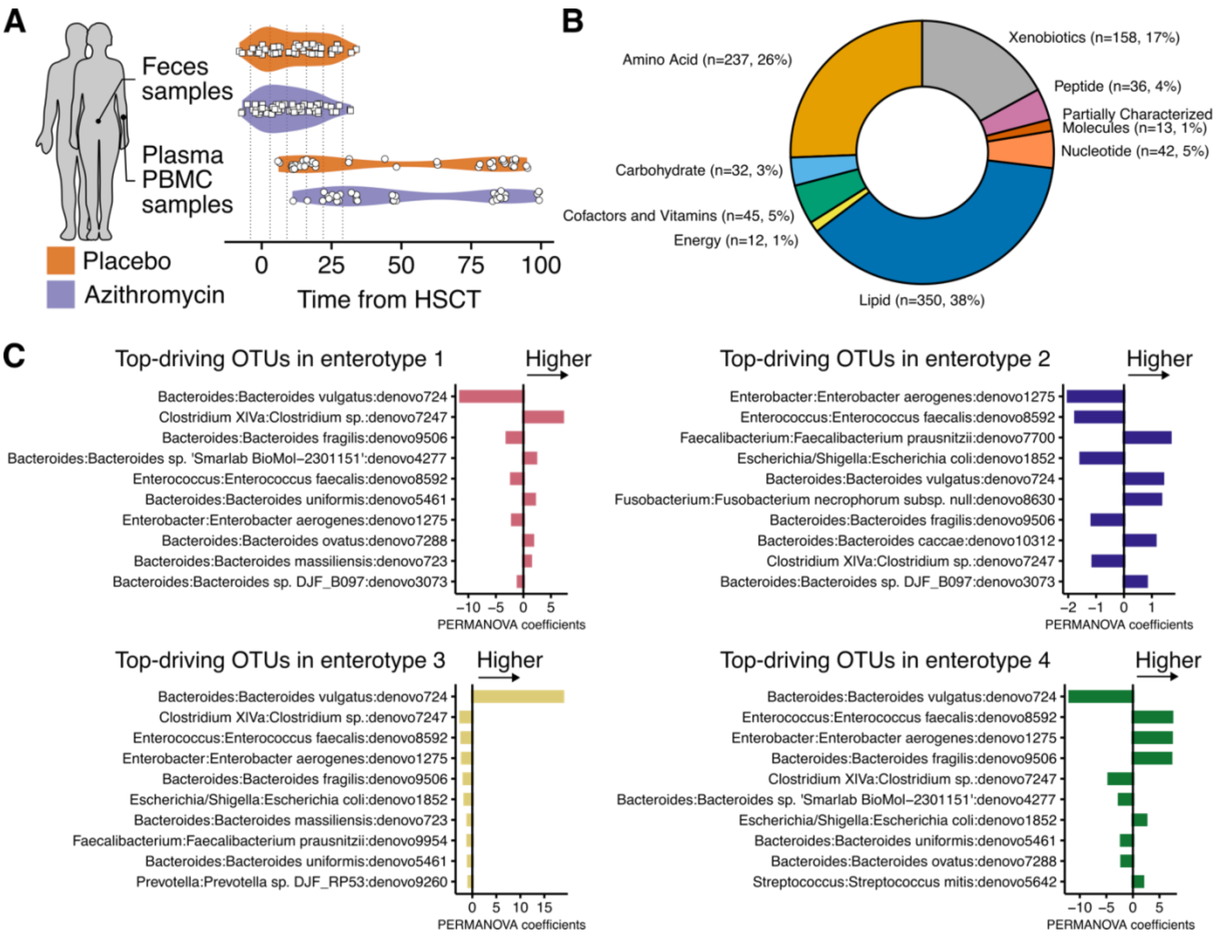


Figure S8. Supplemental material data, related to STAR Methods. A.) Violin plots depicting time from allogeneic hematopoietic stem cell transplantation (HSCT) to feces and blood samples collection. B.) Metabolites analyzed according to the corresponding superpathways. Numbers surrounding the donut plots represent the number of metabolites studied within each superpathway. C.) Top enterotypes-driving taxa. Bar plots exhibiting top 10 absolute permutational multivariate analysis of variance (PERMANOVA) coefficient of phylotypes (OTU) in each enterotype.

Table S1. Characteristics of patients included in the present study, related to Figure 1

	All patients	Placebo	Azithromycin	
n=	55 (148 samples)	28 (75 samples)	27 (73 samples)	p-value
Gender, n Female (%)	19 (35)	11 (39)	8 (30)	0.50
Age (years)	55 (40-62)	53 (40-60)	56 (41-64)	0.40
Diagnosis, n (%)				
Acute myeloid leukemia	11 (20)	3 (11)	8 (30)	0.08
Acute lymphoid leukemia	9 (16)	6 (21)	3 (11)	
Myelodysplastic neoplasm	13 (24)	8 (29)	5 (19)	
Non-Hodgkin lymphoma	9 (16)	7 (25)	2 (7.4)	
Hodgkin lymphoma	1 (1.8)	0 (0)	1 (3.7)	
Chronic lymphoid leukemia	3 (5.5)	0 (0)	3 (11)	
Chronic myeloid leukemia	1 (1.8)	1 (3.6)	0 (0)	
Other malignancies	8 (15)	3 (11)	5 (19)	
Disease risk index, (%)				
Low	2 (3.6)	0 (0)	2 (7.4)	0.50
Intermediate	22 (40)	12 (43)	10 (37)	
High	31 (56)	16 (57)	15 (56)	
Disease status at transplant, (%)				
CR1	10 (18)	5 (18)	5 (19)	0.99
>CR1	19 (35)	10 (36)	9 (33)	
Other	26 (47)	13 (46)	13 (48)	
Donor type, n (%)				
Related	21 (38)	12 (43)	9 (33)	0.50
Unrelated	34 (62)	16 (57)	18 (67)	
CMV serology (Donor/Recipient), n (%)				
-/-	20 (36)	11 (39)	9 (33)	0.80
-/+	12 (22)	7 (25)	5 (19)	
+/-	8 (15)	3 (11)	5 (19)	
+/+	15 (27)	7 (25)	8 (30)	
Source of stem cells, n (%)				
Peripheral blood	51 (93)	25 (89)	26 (96)	0.70
Bone marrow	2 (3.6)	2 (7.1)	0 (0)	
Cord blood	2 (3.6)	1 (3.6)	1 (3.7)	
Conditioning regimen, n (%)				
Myeloablative	16 (29)	8 (29)	8 (30)	0.99
Non-myeloablative	39 (71)	20 (71)	19 (70)	
Graft-vs-host disease prophylaxis, n (%)				
Ciclosporin-methotrexate	15 (27)	7 (25)	8 (30)	0.99
Ciclosporin-mycophenolate mofetil	38 (69)	20 (71)	18 (67)	
Post-transplant cyclophosphamide	0 (0)	0 (0)	0 (0)	
Other	2 (3.6)	1 (3.6)	1 (3.7)	
Acute GVHD, n (%)				
aGVHD	35 (64)	16 (57)	19 (70)	0.30
Death before aGVHD	11 (20)	5 (18)	6 (22)	
No event at last follow up	9 (16)	7 (25)	2 (7.4)	
Chronic GVHD, n (%)				
Mild	16 (29)	6 (21)	10 (37)	0.2
Moderate	11 (20)	7 (25)	4 (15)	
Severe	2 (3.6)	0 (0)	2 (7.4)	
No event at last follow up	26 (47)	15 (53)	11 (41)	
Bronchiolitis obliterans syndrome, n (%)				
Yes	1 (1.8)	0 (0)	1 (3.7)	NA
Relapse at 12 months, n (%)				
Relapse	15 (27)	5 (22)	10 (38)	0.20
Death before 12 months	6	5	1	
No event at 12 months	34 (69)	18 (78)	16 (62)	
Nutrition, n (%)				
Enteral	8 (15)	4 (15)	4 (15)	0.40
Oral	32 (60)	14 (52)	18 (69)	
Intraveinous	13 (25)	9 (33)	4 (15)	
Unknown	2	1	1	
Antibiotics, (%)				
Amoxicilin	53 (96)	28 (100)	25 (93)	0.20
Ofloxacin	53 (96)	28 (100)	25 (93)	0.20
Piperacilline-Tazobactam	39 (71)	20 (71)	19 (70)	0.99
Ceftazidim	16 (29)	9 (32)	7 (26)	0.60
Vancomycin	24 (44)	14 (50)	10 (37)	0.30
Imipenem Cilastin	9 (16)	4 (14)	5 (19)	0.70
Amikacin	48 (87)	24 (86)	24 (89)	0.99
Ceftriaxone	1 (1.8)	0 (0)	1 (3.7)	0.50
Cephazoline	11 (20)	8 (29)	3 (11)	0.11
Sulfamethoxazole Trimethoprim	55 (100)	28 (100)	27 (100)	-
Levofloxacin	1 (1.8)	1 (3.6)	0 (0)	0.99
Metronidazole	1 (1.8)	1 (3.6)	0 (0)	0.99
Cefotaxime	1 (1.8)	1 (3.6)	0 (0)	0.99
Ganciclovir	4 (7.3)	3 (11)	1 (3.7)	0.60
Valaciclovir	54 (98)	28 (100)	26 (96)	0.50
Fluconazole	13 (24)	6 (21)	7 (26)	0.70
Caspofungine	10 (18)	7 (25)	3 (11)	0.30
Micafungine	1 (1.8)	1 (3.6)	0 (0)	0.99
Posaconazol	4 (7.3)	3 (11)	1 (3.7)	0.60
Voriconazole	9 (16)	8 (29)	1 (3.7)	0.025

Continuous variables are described with median and interquartile range and compared with bilateral Wilcoxon test. Fisher's test was used to compare frequencies. GVHD: graft-versus-host disease.

Table S2: Type of nutrition support and antibiotic used during the procedure, related to Figure 1

	All patients	Placebo	Azithromycin	
n=	55 (148 samples)	28 (75 samples)	27 (73 samples)	p-value
Nutrition, n (%)				
Enteral	8 (15)	4 (15)	4 (15)	0.40
Oral	32 (60)	14 (52)	18 (69)	
Intraveinous	13 (25)	9 (33)	4 (15)	
Unknown	2	1	1	
Antibiotics, (%)				
Beta-lactamin				
Amoxicilin	53 (96)	28 (100)	25 (93)	0.20
Piperacilline-Tazobactam	39 (71)	20 (71)	19 (70)	0.99
Glycopeptide				
Vancomycin	24 (44)	14 (50)	10 (37)	0.30
Cephalosporin				
Ceftriaxone	1 (1.8)	0 (0)	1 (3.7)	0.50
Cephazoline	11 (20)	8 (29)	3 (11)	0.11
Ceftazidim	16 (29)	9 (32)	7 (26)	0.60
Cefotaxime	1 (1.8)	1 (3.6)	0 (0)	0.99
Quinolone				
Ofloxacin	53 (96)	28 (100)	25 (93)	0.20
Levofloxacin	1 (1.8)	1 (3.6)	0 (0)	0.99
Carbapenem				
Imipenem Cilastin	9 (16)	4 (14)	5 (19)	0.70
Sulfonamides				
Sulfamethoxazole Trimethoprim	55 (100)	28 (100)	27 (100)	NA
Nitroimidazole				
Metronidazole	1 (1.8)	1 (3.6)	0 (0)	0.99
Aminoglycoside				
Amikacin	48 (87)	24 (86)	24 (89)	0.99
Anti-viral				
Ganciclovir	4 (7.3)	3 (11)	1 (3.7)	0.60
Valaciclovir	54 (98)	28 (100)	26 (96)	0.50
Anti-fungal				
Fluconazole	13 (24)	6 (21)	7 (26)	0.70
Caspofungine	10 (18)	7 (25)	3 (11)	0.30
Micafungine	1 (1.8)	1 (3.6)	0 (0)	0.99
Posaconazol	4 (7.3)	3 (11)	1 (3.7)	0.60
Voriconazole	9 (16)	8 (29)	1 (3.7)	0.025

Continuous variables are described with median and interquartile range and compared with bilateral Wilcoxon test. Fisher's test was used to compare frequencies. GVHD: graft-versus-host disease.

Table S3. Clinical characteristics of patients according to enterotype trajectories (among patients with at least two samples studied), related to Figure 2

	Changed enterotype	Identitical enterotype	
n patients=	22	16	p-value
Gender, n Female (%)	5 (23%)	6 (38%)	0.50
Age (years)	57 (46-63)	54 (41-63)	0.7
Group of treatment, n (%)			
Placebo	13 (59%)	6 (38%)	0.2
Azithromycin	9 (41%)	10 (62%)	
Diagnosis, n (%)			
Acute myeloid leukemia	4 (18%)	5 (31%)	0.50
Acute lymphoid leukemia	4 (18%)	3 (19%)	
Myelodysplastic neoplasm	7 (32%)	1 (6.2%)	
Non-Hodgkin lymphoma	2 (9.1%)	3 (19%)	
Hodgkin lymphoma	1 (4.5%)	0 (0%)	
Chronic lymphoid leukemia	1 (4.5%)	2 (12%)	
Chronic myeloid leukemia	1 (3.6)	0 (0)	
Other malignancies	2 (9.1%)	3 (19%)	
Disease status at transplant, (%)			
CR1	3 (14%)	6 (38%)	0.30
>CR1	8 (36%)	5 (31%)	
Other	11 (50%)	5 (31%)	
Donor type, n (%)			
Related	7 (32%)	7 (44%)	0.50
Unrelated	15 (68%)	9 (56%)	
CMV serology (Donor/Recipient), n (%)			
-/-	5 (23%)	4 (25%)	>0.9
-/+	5 (23%)	3 (19%)	
+/-	4 (18%)	3 (19%)	
+/+	8 (36%)	6 (38%)	
Source of stem cells, n (%)			
Peripheral blood	20 (91%)	14 (88%)	>0.9
Bone marrow	1 (4.5%)	1 (6.2%)	
Cord blood	1 (4.5%)	1 (6.2%)	
Conditioning regimen, n (%)			
Myeloablative	5 (23%)	5 (31%)	0.70
Non-myeloablative	17 (77%)	11 (69%)	
Graft-vs-host disease prophylaxis, n (%)			
Ciclosporin-methotrexate	3 (14%)	6 (38%)	0.15
Ciclosporin-mycophenolate mofetil	17 (77%)	10 (62%)	
Post-transplant cyclophosphamide	0 (0)	0 (0)	
Other	2 (9.1%)	0 (0%)	
Acute GVHD, n (%)			
aGVHD	14 (64%)	8 (50%)	0.30
Death before aGVHD	3 (14%)	6 (38%)	
No event at last follow up	5 (23%)	2 (12%)	
Chronic GVHD, n (%)			
Mild	7 (58%)	7 (78%)	0.70
Moderate	3 (25%)	2 (22%)	
Severe	2 (17%)	0 (0%)	
No event at last follow up	10	7	
Bronchiolitis obliterans syndrome, n (%)			
Yes	1 (100%)	0 (0%)	NA
Relapse at 12 months, n (%)			
Relapse	4 (19%)	8 (57%)	0.020
Death before 12 months	1	2	
No event at 12 months	17 (81%)	6 (43%)	
Nutrition, n (%)			
Enteral	5 (23%)	1 (6.7%)	0.30
Oral	13 (59%)	9 (60%)	
Intraveinous	4 (18%)	5 (33%)	
Unknown	0	1	
Antibiotics, (%)			
Beta-lactamin			
Amoxicilin	21 (95%)	15 (94%)	>0.9
Piperacilline-Tazobactam	15 (68%)	12 (75%)	0.7
Glycopeptide			
Vancomycin	9 (41%)	6 (38%)	0.8
Cephalosporin			
Ceftriaxone	1 (4.5%)	0 (0%)	>0.9
Cephazoline	4 (18%)	4 (25%)	0.7
Ceftazidim	6 (27%)	3 (19%)	0.7
Cefotaxime	0 (0%)	0 (0%)	NA
Quinolone			
Ofloxacin	21 (95%)	16 (100%)	>0.9
Levofloxacin	1 (4.5%)	0 (0%)	>0.9
Carbapenem			
Imipenem Cilastin	2 (9.1%)	2 (12%)	>0.9
Sulfonamides			
Sulfamethoxazole Trimethoprim	22 (100%)	16 (100%)	
Nitroimidazole			
Metronidazole	0 (0%)	0 (0%)	NA
Aminoglycoside			
Amikacin	20 (91%)	14 (88%)	>0.9
Anti-viral			
Ganciclovir	2 (9.1%)	2 (12%)	>0.9
Valaciclovir	21 (95%)	16 (100%)	>0.9
Anti-fungal			
Fluconazole	7 (32%)	2 (12%)	0.3
Caspofungine	3 (14%)	3 (19%)	0.7
Micafungine	0 (0%)	0 (0%)	NA
Posaconazol	1 (4.5%)	1 (6.2%)	>0.9
Voriconazole	5 (23%)	2 (12%)	0.7

Continuous variables are described with median and interquartile range and compared with bilateral Wilcoxon test. GVHD: graft-versus-host disease.